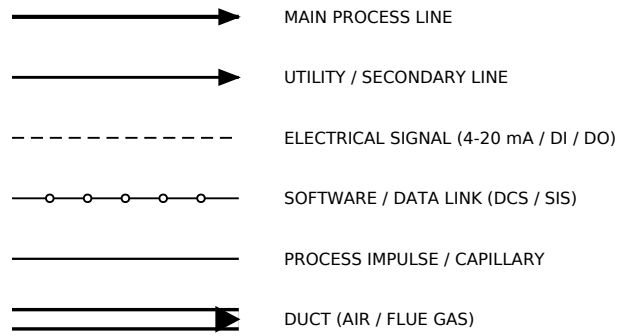
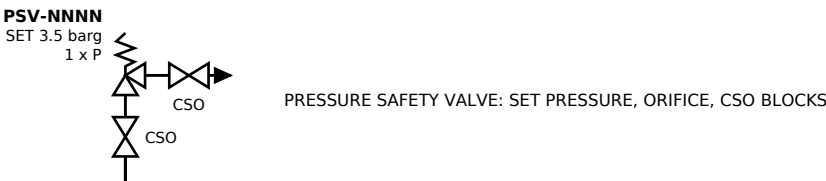
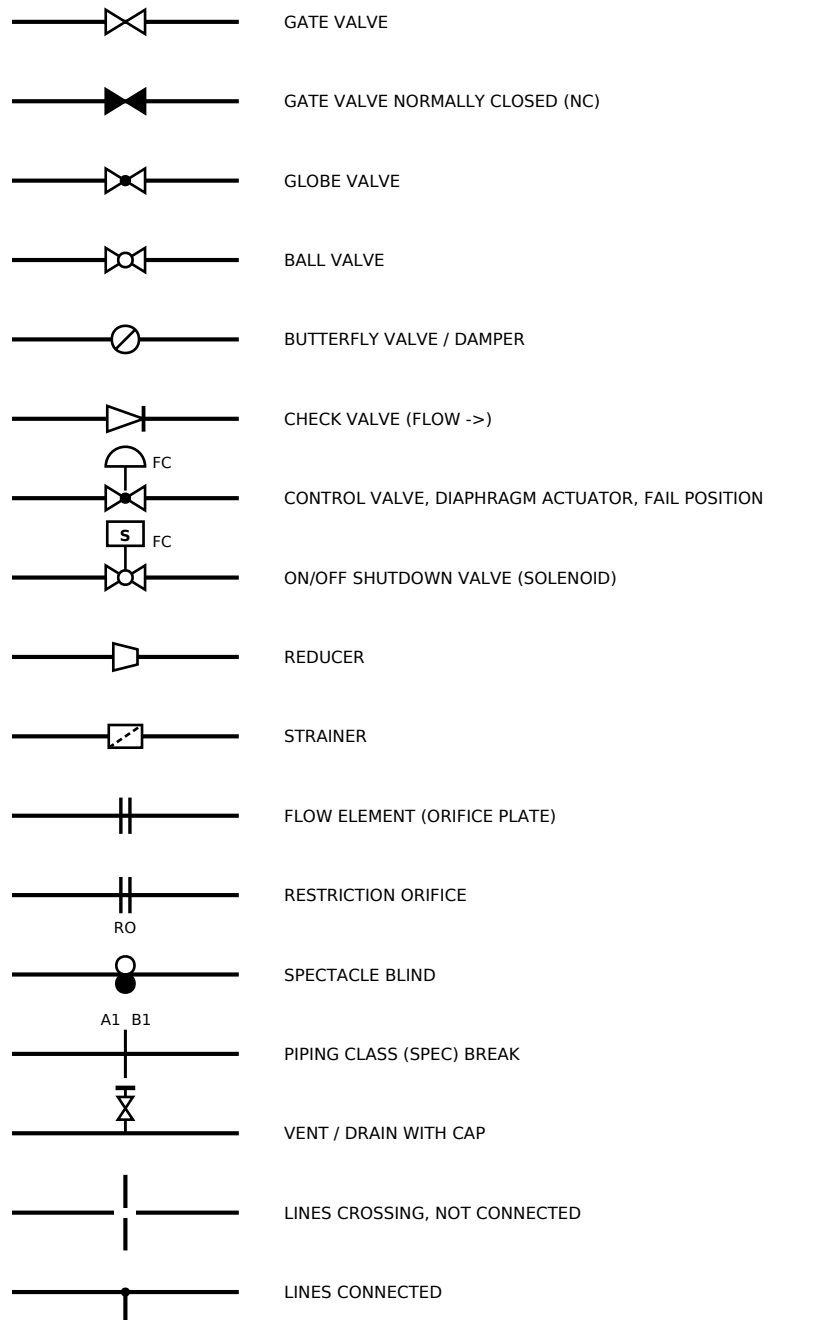


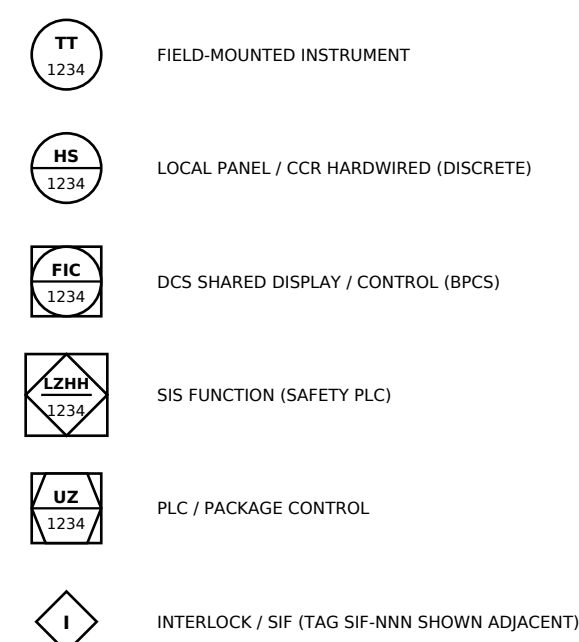
LINE SYMBOLS



VALVES & PIPING COMPONENTS



INSTRUMENT SYMBOLS (ISA-5.1)



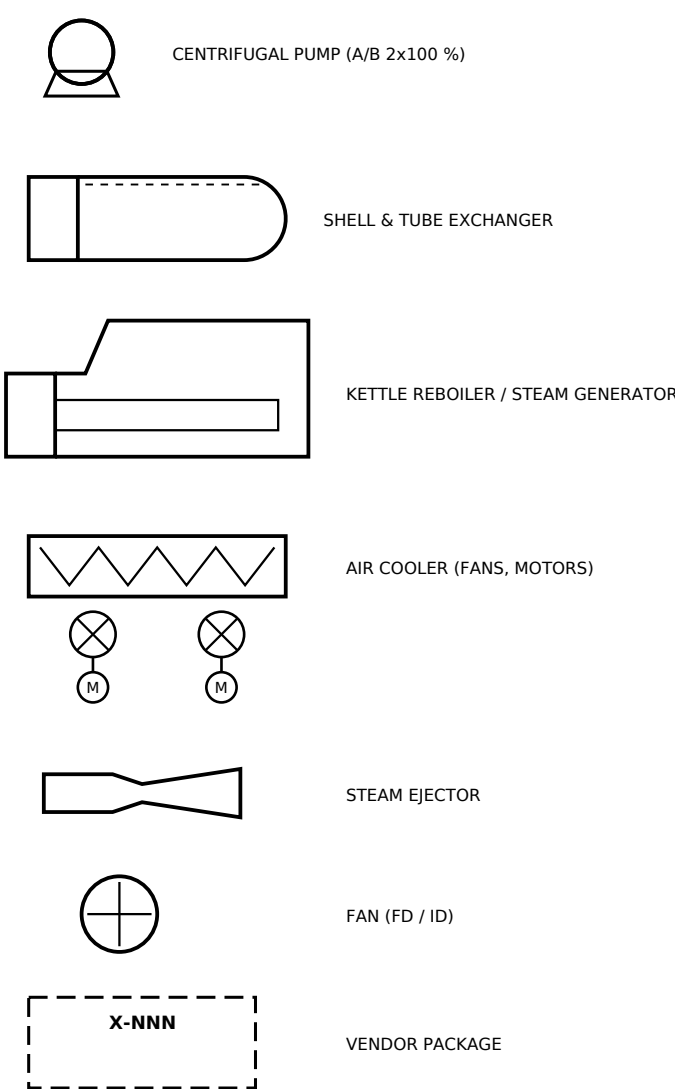
INSTRUMENT IDENTIFICATION LETTERS

A ANALYSIS (O₂, pH) B BURNER / FLAME
F FLOW H HAND
L LEVEL P PRESSURE
T TEMPERATURE X/Z ON-OFF / POSITION
PD / TD DIFFERENTIAL FF FLOW RATIO
Succeeding: E element, G glass, I indicate, C control,
T transmit, V valve, Y relay/compute, S switch, Z (2x4) SIS,
AH/AH/ALL alarms; A/B/C suffix = redundant SIS

ABBREVIATIONS

FC / FO / FL FAIL CLOSED / OPEN / LAST
CSO / CSC CAR-SEALED OPEN / CLOSED
NC / NO NORMALLY CLOSED / OPEN
NMF NORMALLY NO FLOW
TPA / MPA / BPA TOP / MID / BOTTOM PUMP/AROUND
SS STRIPPING STEAM
DBB DOUBLE BLOCK & BLEED
ROSOV / EIV REMOTE OPERATED SHUT-OFF VALVE
COI / COT COIL OUTLET / INLET TEMP.
RO RESTRICTION ORIFICE
FA FLAME ARRESTOR
OSBL OUTSIDE BATTERY LIMITS
BMS BURNER MANAGEMENT SYSTEM
2o3 TWO-OUT-OF-THREE VOTING

EQUIPMENT SYMBOLS



LINE NUMBER

12"-P-100-002-B1-H

NPS (in) - FLUID - AREA - SEQUENCE - PIPING CLASS - INSULATION
AREA: 100 CDU, 200 VDU, 900 UNIT UTILITIES

FLUID CODES

P Process hydrocarbon
PG Process gas / vapour
SW Sour water
WW Wash water / brine
LS LP steam
HS MP steam
HS HP steam
CD Condensate
BFW Boiler feed water

CWS Cooling water supply
CWR Cooling water return
FG Fuel gas
FL Flare
BD Blowdown / drain
IA Instrument air
N Nitrogen
CH Chemical

INSULATION CODES

H HEAT CONSERVATION
P PERSONNEL PROTECTION
ST STEAM TRACED + INSULATED
C COLD
N NONE

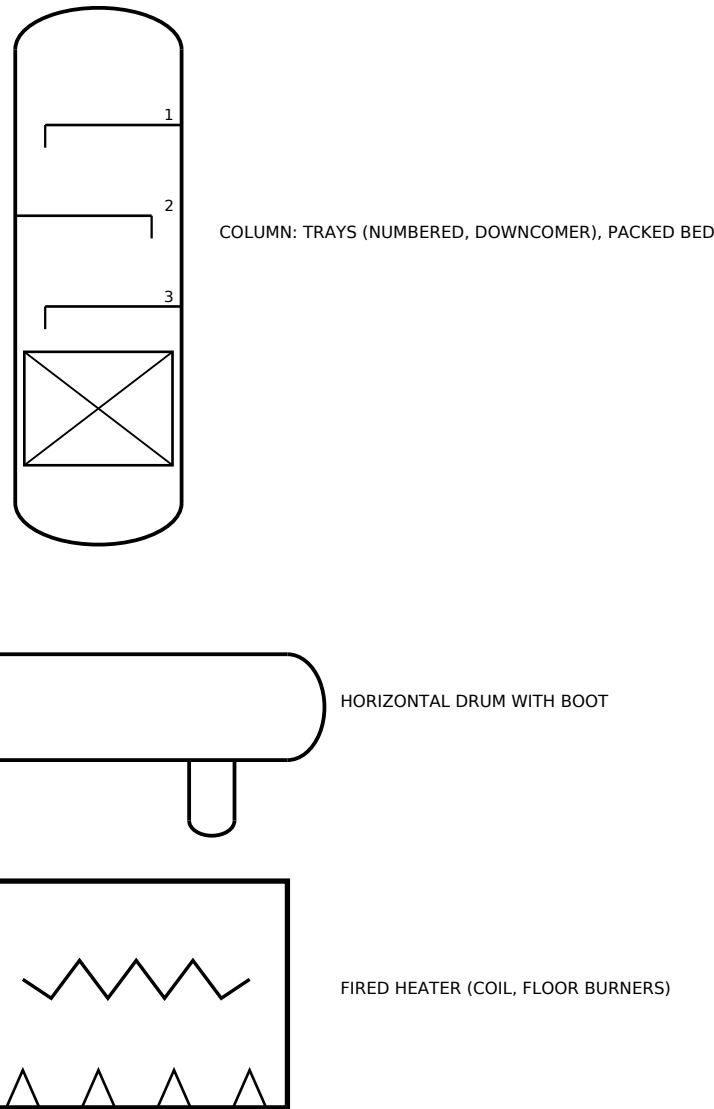
PIPING CLASSES (SUMMARY, SEE CFU-000-PI-SPC-001)

A1 150# Carbon steel ASTM A106 Gr.B
A2 150# Carbon steel, HIC-resistant (NACE MR0103)
B1 300# Carbon steel ASTM A106 Gr.B
B2 300# SC-1/2Mo ASTM A335 P5
B3 300# 9Cr-1Mo ASTM A335 P9
C1 600# Carbon steel, killed ASTM A106 Gr.B
A3 300# Carbon steel, HIC-resistant (NACE MR0103)
S1 150# Carbon steel ASTM A106 Gr.B
S2 600# 1.25Cr-1/2Mo ASTM A335 P11
S3 300# Carbon steel ASTM A106 Gr.B
F1 150# Carbon steel, killed, impact-tested (A33)
U1 150# Carbon steel (cement-lined for CW >= 12°)
U2 150# SS304 (A) / galvanised CS (N2)
DRAWING INDEX A335 P5

Hydrocarbon <= 230 C, general
Sour water, OH vapour/condensate, wash water/brine
Hydrocarbon <= 260 C, pump discharges
Sulfidic hydrocarbon 260-400 C (incl crude, AR, BPA, HvGO, VR)
Heater outlets / transfer lines (H-101, H-201) and > 400 C
LPG / stabiliser, BFW > 300#
Wash water / brine / sour water above 150# rating (desalter pr
LP / MP steam, condensate
HP steam 41 barg / 400 C
MP steam 10.3 barg / 250-290 C design (exceeds S1)
Flare / relief headers, -29 to 350 C
Cooling water, utility water, LP BFW
Instrument air, nitrogen
Vacuum vapour lines (C-201 OH); transfer line > 400 C -> B3

LEGEND, SYMBOLS & ABBREVIATIONS
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN
ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS
HOT PREHEAT TRAIN E-106 TO E-111 & E-113
ATMOSPHERIC HEATER H-101 - PROCESS COILS
ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM
ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMP/AROUNDS
SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN
C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102
NAPHTHA STABILISER C-105
NAPHTHA SPLITTER C-106
VACUUM HEATER H-201
VACUUM COLUMN C-201
VACUUM PUMP/AROUNDS & PRODUCTS
VACUUM EJECTOR SYSTEM & HOTWELL D-201
FUEL GAS, FLARE KO & UTILITY HEADERS

EQUIPMENT DATA BOX



NOTES
1. Legend applies to all P6&Ds CFU-xxx-PR-PID-001 to -016.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM LEGEND, SYMBOLS & ABBREVIATIONS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	1 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-PR-PID-000			A